Behaviour-Aware Urgency-Sensitive Multilingual Voice-Based Local Service Discovery and Workforce Empowerment System Using a Multi-Agent AI Framework

R26-IT-109

Project Proposal Report

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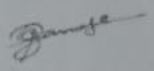
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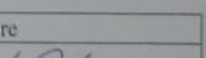
March 2026

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

People now use online service platforms to access local services because these platforms have experienced rapid growth. Existing systems do not provide support for natural communication patterns and accessibility requirements which are essential in multilingual environments like Sri Lanka where users switch between Sinhala and English. The digital platforms of various local service providers such as plumbers and electricians and domestic workers do not exist because these workers lack technical skills and face difficulties in completing the registration process.

This research presents a local service discovery system which utilizes multilingual voice recognition together with behaviour-driven intelligence and multi-agent artificial intelligence technology. The system enables service seekers and providers to interact through natural voice communication which supports multiple languages while they handle urgent requests which enhance accessibility and reliability and inclusive access to services.

Multiple intelligent components exist within the solution. The multilingual voice intent model processes code-switching and incomplete speech while detecting urgency through acoustic signals which include pitch and speech rate and energy level variations. The system uses a cooperative multi-agent coordination layer which determines service provider locations and matching service demand needs and calculates their operational capacity and dependability and their urgent service requests. The AI-driven workforce intelligence module establishes fair job allocation methods while managing worker reputation and predicting employment needs to create sustainable job opportunities.

The system offers non-technical providers a voice-based onboarding system which requires no technical skills for use. Providers can create accounts and develop their profiles and use the platform by engaging in voice conversations which require no text writing or password usage. The framework supports multilingual voice prompts which use voice biometric authentication and OTP-based authentication methods through an HTML interface.

Keywords: Multilingual Voice Interaction, Multi-Agent System, Service Discovery, Behaviour-Aware AI, Workforce Intelligence, Voice-Based Authentication, Digital Inclusion.

ACKNOWLEDGEMENT

This research proposal could not have taken place without guidance, support and encouragement of various individuals and institutions.

The research supervisor, Professor Samantha Rajapaksa, deserves special thanks because of the great advice, professional skills, and positive feedback that he offered to me during the process of this proposal development. His support and academic insights were of much help in influencing the direction and quality of this research.

It is also with a lot of sincerity that I would like to thank the co-supervisor Ms. Poorna Panduwawala who provided me with helpful advice, thoughtful suggestions and constant encouragement in preparing this piece of research work.

It is also hoped that the academic staff of the Department of Information Technology at the Sri Lanka Institute of Information Technology (SLIIT) are also similarly appreciated in this regard because without the academic environment, knowledge, and resources that they offered, this proposal would not have materialized.

Lastly, the family members and friends are also recognized due to their encouragement, patience, and support during the preparation of this research proposal.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
ML	Machine Learning
NLP	Natural Language Processing
MFCC	Mel-Frequency Cepstral Coefficients
ASR	Automatic Speech Recognition
LSTM	Long Short-Term Memory
API	Application Programming Interface
SDG	Sustainable Development Goal
HR	Human Resource
IoT	Internet of Things
SLIIT	Sri Lanka Institute of Information Technology
SUS	System Usability Scale
WBS	Work Breakdown Structure
AWS	Amazon Web Services
ARIMA	Autoregressive Integrated Moving Average
XGBoost	Extreme Gradient Boosting
MAE	Mean Absolute Error
RMSE	Root Mean Square Error

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1. INTRODUCTION

1.1 Background and Context

Such increased pace of development of digital platforms and service-oriented mobile applications has contributed greatly to the changes in modern labour markets and employment patterns. Most industries have adopted platform-based ecosystems of services whereby independent employees offer services like technical help, transportation, house upkeep, and expert service utilization through online platforms. These websites have provided the flexibility of employment and enhanced accessibility of services to the users. Because of this, the employment forms based on gigs have become a significant part of the contemporary digital economy, especially in the service-oriented industry where the availability of on-demand workforce is crucial.

Nevertheless, in spite of these innovations, the platform-based employment systems continue to experience a number of threats associated with the management of the workforce, equity in job distribution, and job maintenance. In most digital service platforms, work placements are mostly made on the basis of basic algorithmic processes which rank these factors which include proximity, availability or available reputation scores. Although these methods can be effective in automatizing the process of matching services they do not pay much attention to the fair allocation of tasks among the employees or, the sustainability of job prospects. This means that a good share of the available tasks can be allocated to some service providers and is underutilised in the case of others who have the relevant skills and availability.

Such problems are especially relevant to developing economies where a large part of the population depends on gig-based or other informal jobs as their main source of income. The International Labour Organization (ILO, 2023) estimates that around 60 percent of the world labour force works on informal jobs with the rate being over 80 percent in some countries of South Asia. In Sri Lanka in particular, the Department of Census and Statistics (2023) approximates that more than 65 per cent of the employment is informal, and that some considerable part is reliant on demand-based services. It is estimated that the global gig economy is growing by USD 455 billion in 2023 to more than USD 1.8 trillion in 2032 (Statista, 2024), which means that there is a strong need to develop smart systems that can accommodate this underprovided and rapidly increasing workforce. In the absence of smart monitoring of the workforce, it is hard to determine where there is a lack of skills, trends in

demand of the services or even have a balanced distribution of tasks among the service providers.

Current digital service platforms have been more interested in service discovery and transactions, as opposed to workforce wisdom and labour sustainability. The traditional platforms use a lot of reputation systems that are not dynamic and simple ranking algorithms to decide on the assignment of tasks to workers. These systems usually appraise the workers with cumulative scores or past performance without referring to the new development in the service quality. In addition, the majority of platforms do not have predictive analytics features that could be used to determine the level of demand and supply in the workforce, predict shortages of skills, or analyse the financial sustainability of gig workers over the long term.

To cope with these difficulties, this study suggests an AI-Driven Employment Empowerment and Workforce Intelligence Framework that is the Component 3 of the entire system. The proposed framework proposes a workforce intelligence layer that is data-based to enhance the sustainability of employment and efficiency of service platforms. The framework consists of five major modules, namely; Intelligent employee bank, Fairness-Aware dynamic task allocation mechanism, Dynamic reputation evolution model, Workforce Demand and Income Stability prediction modules.

This study is supported by the United Nations Sustainable Development Goal (SDG) 8: Decent Work and Economic Growth, which highlights the importance of productive labour market and equitable working environments in contemporary labour market. SDG 9: Industry, Innovation, and Infrastructure is another area of contribution made by the research, which encourages the development of smart digital platforms that will combine AI and data-driven technology in managing work forces. In terms of the academic field, the proposed work belongs to the cluster of the Artificial Intelligence and Data Science research which is the workforce analytics and intelligent decision-making systems.

1.2 Literature Review

The emergence of intelligent workforce management systems to solve sophisticated employment issues in digital service platforms has been made possible by recent progress in the artificial intelligence, machine learning, and predictive analytics. The studies in this area are becoming more geared towards employee profiling, automated task assignment, trust and reputation modeling, workforce demand prediction, and income stability prediction of gig workers. Nevertheless, current systems continue to focus mainly on efficiency in operations services, as opposed to equity in employment, sustainability in the workforce, and empowerment of workers.

1.2.1 Artificial Intelligence driver profiling and talent development.

Employee profiling and talent analytics brought about by the application of artificial intelligence and machine learning have become increasingly discussed within the scholarly community over the last several years. As it was established by Upadhyay and Khandelwal (2018), AI-driven human resource management systems may considerably enhance the outcomes of recruitment and employee development by using massive datasets of workforce. An extensive overview of AI applications in talent analytics explored the prospects of how machine learning models can be used to analyse employee profiles, skills history, and labour market data to determine workforce trends and skill gaps [1]. It is a survey based on the data of employment of more than 500 million professional accounts, which demonstrated the possibilities of predictive modelling to facilitate organized workforce intelligence at scale.

Based on this, Kulkarni and Che (2025) explored the potential of AI-based personalisation on human resource management to implement personalised training programmes, detect individual skill gaps, and provide continuous support to align the capabilities of workers to the dynamics of organisational needs [2]. Employee profiling based on AI is already implemented in the existing commercial platforms like Linked In Talent Insights, IBM Watson Talent, and SAP SuccessFactors. An example here would be LinkedIn Talent Insights, which provides real-time labour market and workforce planning dashboards and is prohibitively expensive to small operators. IBM Watson Talent is used to filter vast numbers of candidates in structured enterprise pipelines using natural language processing. SAP SuccessFactors has been offering competency mapping and succession planning as part of the formal payroll and HR management. Nevertheless, all of the three solutions are mostly suited to large formal businesses with in place employment records and standardised information and none of them talks about informal or gig-based service workers in developing economies. One of the main weaknesses is that they use formal employment records which are not

readily available in informal service sector to encourage the Intelligent Employee Bank presented in this study.

1.2.2 Digital Marketplace Fairness With Appropriate Task Allocation

Platform-based service ecosystems are centered around algorithmic distribution of tasks and have important consequences to employment equity. A survey based on the Organisational Justice Theory and the survey among 435 Uber drivers showed that perceived fairness of the algorithmic HR decisions is a significant determinant of worker satisfaction and faith in the platform systems [4]. A review on the use of algorithms in managing the gig economy identified a systematic review of the role of AI algorithms in the coordination of the work of platforms like Uber, Airbnb, and Fiverr [5]. The Uber dispatch algorithm, such as that, favours distance to drivers and information about surge pricing but does not offer any solutions to evenly distributing earnings among its drivers. The ranking system of Fiverr also focuses the visibility and task opportunities on sellers that have a proven history of reviews, thus establishing a self-reinforcing focus of income on the highly rated freelancers. The survey found that these systems of allocation tend to favor the disadvantaged workers who are new or those who are geographically remote. Studies on decentralised scheduling of tasks between multi-agents proposed a two-layer framework with envy-minimisation system models to achieve more equitable allocation of indivisible resources among agents [3].

Regardless of these improvements, the majority of fairness-conscious allocation models have been tested in controlled simulation settings, but seldom combine several fairness dimensions together in a single cohesive optimisation model. The proposed system of Fairness-Aware Dynamic Task Allocation as suggested in this study immediately fills this gap.

1.2.3 Dynamic Reputation and Trust Modelling

The elements of reliable service-oriented platforms include trust and reputation management. Traditional systems employed by sites like Amazon and Uber calculate provider ratings based on cumulative averages, which slowly represent new positive results and proportionately reject single negative events. Amazon, in turn, has a system of seller ratings, such as the lifetime average star rating, which can be fixed after a single bad rated transaction only through dozens of positive reviews. The Uber driver rating system is also based on an overall average in the past few trips but it does not provide any contextual distinction between a rating on an urgent or routine trip and it does not have a formal recovery mechanism of a driver who has fallen short of performance. Alharbi et al. (2024) conducted research to

propose agent-based decentralised trust in smart IoT service environments, and the study has shown that the multi-context quality-of-service evaluation enhances the accuracy and fairness of provider selection [6]. The prototype dynamic service selection model suggested by Zheng et al. (2011) provided fundamental principles of reputation-based provider ranking with dishonesty reporting and time-related trust decays [7], is still considered a classic reference since no more recent united frameworks with the same principle can be found. The Dynamic Reputation Evolution Model used in this study will deal with the inability of existing systems to implement a time-weighted score that allows recovering the previous negative ratings.

1.2.4 Workforce Demand Prediction and Labour Market Forecasting

Workforce planning analytics is an emerging field of application. As Singapore et al. (2024) showed, machine learning models with gradient boosting and LSTM-based architectures always performed better than the classical statistical models in forecasting the near-term changes in labour demand [8]. The systematic review of AI methods in more than 80 studies showed that predictive analytics models have high levels of utility in the attrition of workforce, skill shortages, and staffing decisions [9]. One of the main weaknesses of the current systems is that they rely on big and organized historical data which is usually not accessible on informal gig networks, which is why the proposed module of the Workforce Demand Prediction is needed in this study.

1.2.5 Job Stability and Income Sustenance for Gig Workers

One of the most persistently recorded issues of gig and informal service workers is income instability across the globe. An in-depth industry report has pointed out that the income of gig workers varies greatly according to the demand patterns and algorithmic mobilization choices of the platforms [10]. In a paper, Technology in Society (2025) analyzed the impact of AI-driven employment pattern change on income distribution in the business of gig employment, suggesting that intelligent platform should include income trajectory modelling as an infrastructural element of worker assistance [11]. The current gig companies, including Fiverr, TaskRabbit, and the Sri Lankan system PickMe, have historical earnings dashboards, which show previous transaction summaries, but none of them provides future income projections, personalised financial sustainability advice, or automatic alerts when trends in income portfolios suggest imminent income deficits. This almost total lack of income intelligence tools was the driving force behind the Income Stability Prediction module of this

study, which implements regression-based and time-series machine learning models on the earnings history of individual providers.

1.3 Research Gap

Although there is massive advancement in AI-based workforce intelligence, there is still a lot of weakness in the present employee profiling, task assignment, reputation modeling, workforce requirement forecasting, and earnings stability solutions, especially among the sectors of gig and informal services.

Employee Profiling and Talent Analytics: Current offerings include LinkedIn Talent Insights, IBM Watson Talent, and SAP SuccessFactors that are oriented towards formal enterprise settings. They base on organized records of employment, so they are not quite applicable to informal workers who do gigs. The majority of them do not have real-time skill shortage notifications or individual underutilization detectors [1][2].

Proficiency-Conscious Task Allocation: Fairness Multity-dimensional: Numerous algorithmic allocation algorithms are operationally efficient and Fairness-Ignorant: Most algorithmic allocation systems are operationally efficient, but insensitive to multi-dimensional fairness. Existing frameworks focus on high-rated or centrally positioned employees and can be only assessed in simulations, and they hardly combine various fairness aspects into a single optimisation model [3][4][5].

Dynamic Reputation and Trust Modelling: Traditional systems apply cumulative average in a static manner, do not reward new improvement and does not encourage providers to resume after failures. In the actual gig platform, time-weighted and context-sensitive scoring is still not fully in practice [6][7].

Workforce Demand Prediction: AI-based demand models work with massive structured data that is accessible only in formal businesses. Informal service platforms do not have this infrastructure, and available designs predict on an aggregate level without skills-specific suggestion of workers and a localised demand hotspot warning [8][9].

Forecasting or Stability of Incomes: Present-day gig services only display historical summaries of earnings without foreseeing the future income patterns or giving personalised advice. Extremely, there is a lack of proactive analytics that can assist workers in predicting changes in income or plan skill development and its relation to the demand [10][11].

Component / Sub-Module	Current Limitations	Source	Gap Addressed
Employee Profiling & Talent Analytics	Reliance on formal employment records; lacks real-time skill shortage alerts; unsuitable for informal/gig workers	[1][2]	Supports informal profiles; detects underutilisation; real-time skill shortage insights
Fairness-Aware Task Allocation	Bias toward high-rated/central workers; simulation-only evaluation; limited multi-factor fairness	[3][4][5]	Integrates skill, location, trust score & opportunity distribution in unified optimisation
Dynamic Reputation & Trust	Static cumulative scoring; ignores recent improvements; limited re-engagement incentives	[6][7]	Context-aware, time-weighted trust scoring; adaptive reputation enabling recovery
Workforce Demand Prediction	Requires large structured datasets; aggregate forecasts lacking worker-specific recommendations	[8][9]	Adaptive ML for sparse data; localised demand hotspot alerts & worker-level skill guidance
Income Stability Prediction	Historical earnings only; no predictive analytics; no personalised financial guidance	[10][11]	Predictive income models with recommendations for skill expansion & diversification

Table 1: Research Gap Summary

Table 1 shows that no current system category can meet all five workforce intelligence dimensions which are necessary for fair and sustainable employment in informal service markets. The proposed AI-Driven Employment Empowerment and Workforce Intelligence Framework directly addresses all five dimensions through its unified architecture which represents a new and valuable development to the domain of intelligent workforce management.

2. OBJECTIVE

2.1 Main Objective

To create and assess an AI-Driven Employment Empowerment and Workforce Intelligence Framework, which implements fair task distribution through dynamic reputation assessment and workforce demand forecasting and income stability prediction. The framework will use a multilingual voice-based local service discovery system. The system aims to improve employment opportunities and sustainable services for gig workers in Sri Lanka.

2.2 Specific Objectives

1.We need to create an Intelligent Employee Bank which will provide real-time structured employee data and use predictive analytics to identify both workforce capacity shortages and skill deficits.

2.The Fairness-Aware Dynamic Task Allocation system will improve job distribution by matching skill levels and work locations and employee trust scores and past work opportunities.

3.The Dynamic Reputation Evolution Model establishes a trust score system which utilizes time-sensitive performance data to recover lost trust from providers who show improvement while it punishes providers who continuously fail to meet standards.

4.The Workforce Demand Prediction module uses machine learning forecasting models to determine demand hotspots which will help predict skill shortages and produce service provider skill development pathways.

5.The Income Stability Prediction system uses provider earning history data to create customized recommendations for service providers who want to expand their income through different skills and locations.

6.The integrated framework will be tested through performance metrics which include fairness index and mean absolute error of demand forecasts and trust score convergence rate and System Usability Scale (SUS) score.

3. METHODOLOGY

3.1 Research Approach

The research uses Design Science Research (DSR) as its methodology because this approach enables the creation of new IT artifacts which researchers can assess in specific research environments. The DSR method allows researchers to design and execute their evaluation process for the workforce intelligence framework which they developed as a practical asset that meets theoretical standards. The research process contains five distinct stages which include problem identification and motivation as well as solution design and artefact development and demonstration and evaluation.

3.2 Technical Methods and Algorithms

Intelligent Employee Bank: A structured relational database schema (PostgreSQL) will store provider profiles enriched with skill vectors, availability calendars, and service history. K-Means clustering will identify workforce skill clusters, while Isolation Forest anomaly detection will flag underutilised workers relative to peer groups.

Fairness-Aware Task Allocation: A multi-objective optimisation algorithm will balance four criteria simultaneously: skill-service match score, real-time travel distance (via Google Maps Distance Matrix API), trust score, and historical opportunity distribution index. Pareto-front optimisation and weighted scoring will generate the final allocation ranking.

Dynamic Reputation Evolution Model: An exponential time-weighting formula will compute dynamic trust scores: $\text{Trust}(t) = \alpha \times R(t) + (1-\alpha) \times \text{Trust}(t-1)$, where α is the recency weight and $R(t)$ is the most recent performance rating. Scores will be normalised on a 0–1 scale and recalculated after each completed job.

Workforce Demand Prediction: LSTM neural networks and gradient boosting regression (XGBoost) will be trained on historical service request data, incorporating temporal features. Models will generate 30-day ahead demand forecasts at the service-category and geographic-zone level.

Income Stability Prediction: Ridge regression and ARIMA time-series models will predict individual provider income trajectories over a 90-day horizon. Feature inputs include earnings history, task acceptance rate, service category distribution, and seasonal demand indices.

3.3 Tools and Platforms

The development team has chosen Python version 3.11 as their main programming language because it provides extensive support for machine learning development. The team will use TensorFlow and Keras as their tools for creating LSTM models. The K-Means Isolation Forest and Ridge Regression algorithms will be implemented through Scikit-learn. The development team will implement gradient boosting models using XGBoost. FastAPI will enable the creation of the RESTful backend system. PostgreSQL will function as the main database system. The mobile application will be developed using React Native technology. The team will host all services on AWS EC2 while using RDS for managed database service.

3.4 Data Collection and Ethical Considerations

Research will gather primary data through a structured survey which will target gig service workers and platform users in Sri Lanka with a goal to reach 150 participants. The survey will investigate current employment patterns and income stability perceptions and technology acceptance. The study will use publicly accessible gig economy datasets and artificial service request data as secondary data sources for model development. SLIIT ethical guidelines will govern all procedures for data collection. The researchers will ask for informed consent from all study participants. The research dataset will not contain any personally identifiable information.

3.5 Validation Strategy and Performance Metrics

The framework will be evaluated using: (1) Task Allocation Fairness Index — Gini coefficient of task distribution, target ≤ 0.2; (2) Demand Forecast Accuracy — Mean Absolute Error, target MAE $\leq$ 15% of actual demand; (3) Trust Score Convergence Rate — number of tasks required to reflect performance changes, target ≤ 5 tasks; (4) Income Prediction Accuracy — Root Mean Square Error of 90-day income forecast; (5) System Usability Scale (SUS) score from user testing, target ≥ 75. Baseline comparisons will be conducted against static averaging reputation systems and proximity-only task allocation.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overview

The proposed system is a multi-layered platform which uses artificial intelligence to identify local services and track workforce activities. The system contains four main components which include (1) the Voice Interaction Layer which processes multilingual speech and identifies urgent needs and determines user intentions; (2) the Multi-Agent Coordination Layer which manages the procedures for searching providers and evaluating their suitability and assigning them to tasks; (3) the Workforce Intelligence Layer which handles employee database management and fair task distribution and reputation tracking and demand prediction and financial stability assessment; and (4) the Behaviour-Aware Interaction Layer which includes user-facing functions that enable success forecasting and security measures and pre-service assistance. The complete four-layer architectural design is shown in Figure 2 and Figure 3 presents a complete overview of Component 3.

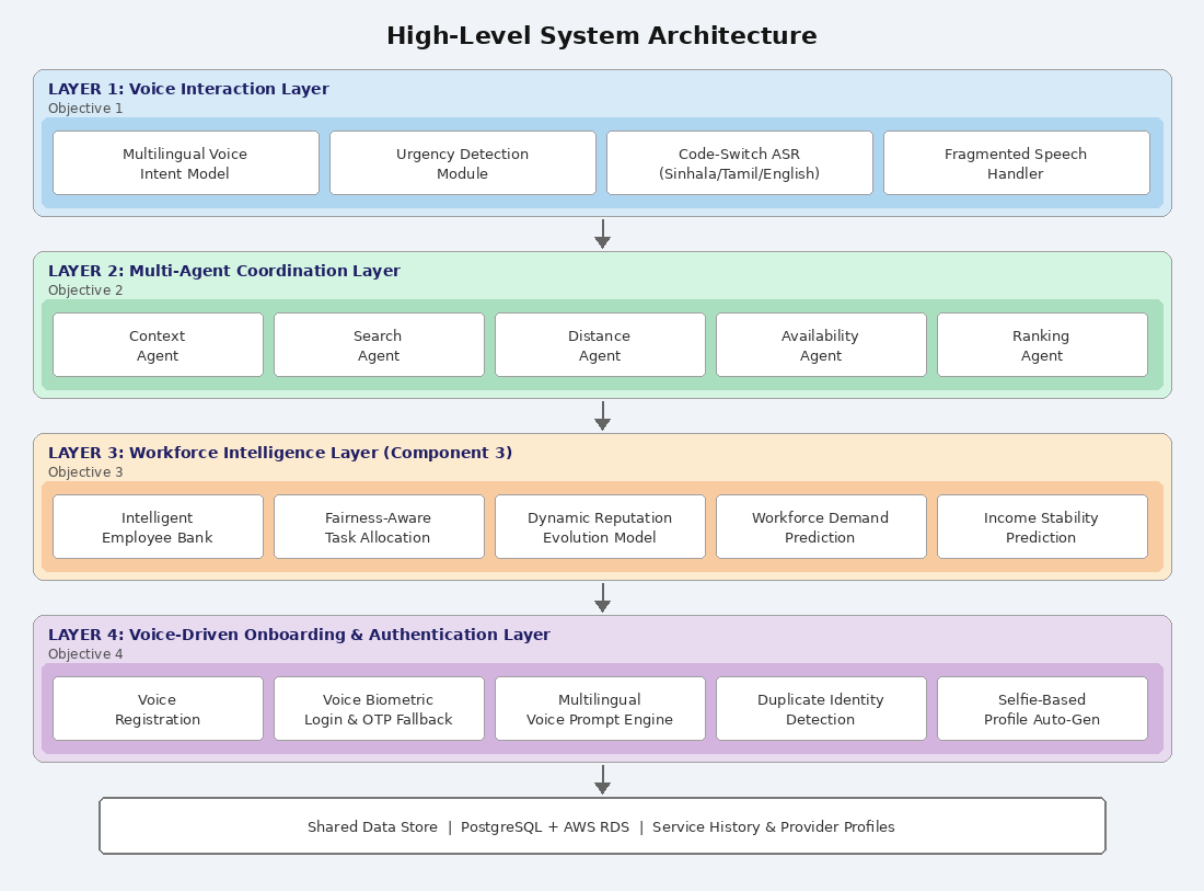


Figure 2: High-Level System Architecture (4 Layers)

4.2 Component 3 — Workforce Intelligence Layer: Detailed Architecture

The Workforce Intelligence Layer is the focus of this component which contains five integrated modules that handle different workforce management and analytics functions. The internal architecture details in Figure 3 display the five modules' interactions with the common data storage system and the controller/API gateway.

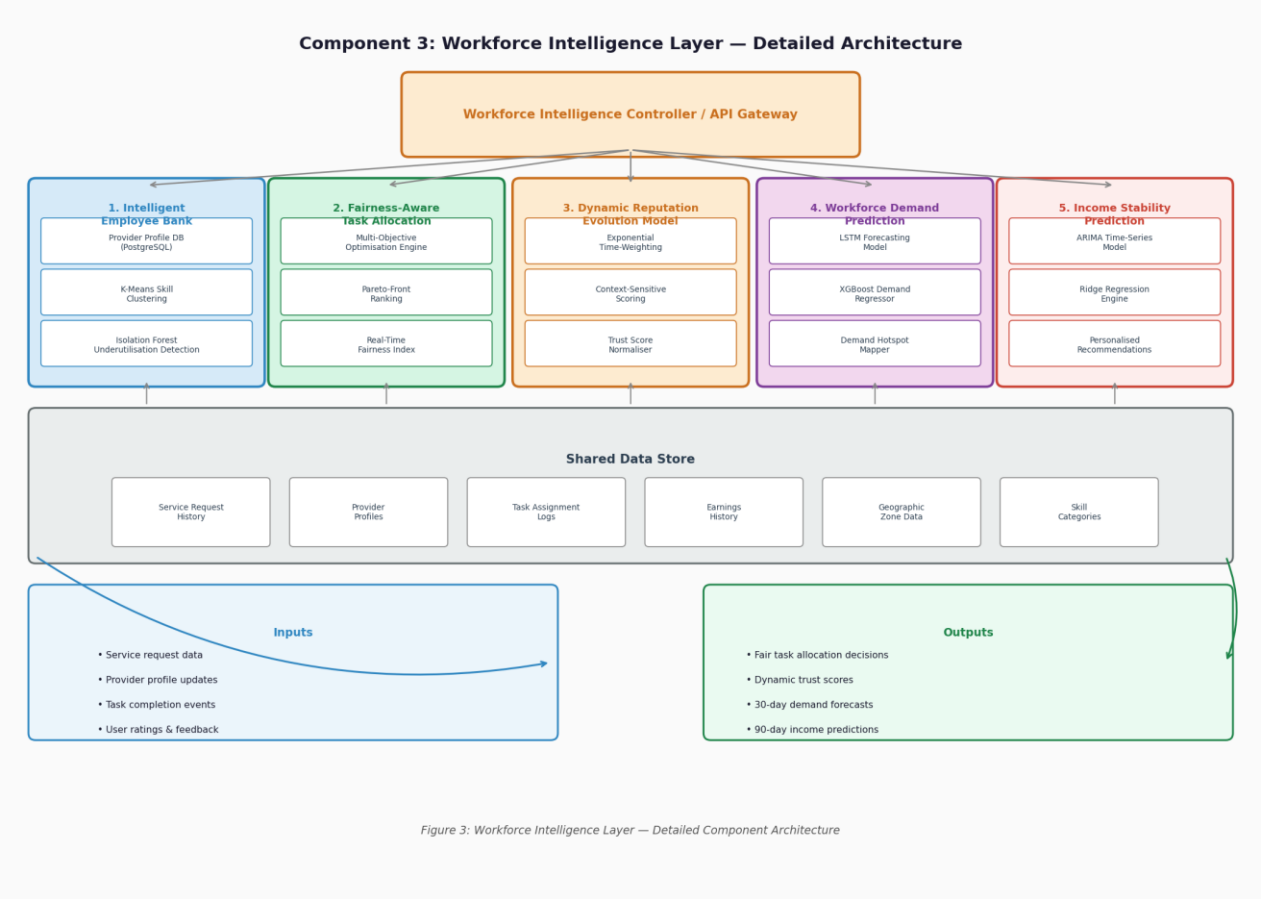


Figure 3: Component 3 — Workforce Intelligence Layer Detailed Architecture

- Intelligent Workers Bank: A PostgreSQL database containing ML-generated skill vectors, user-electivity index values, and workforce-at-risk matrix scores.
- Fairness-Aware Task Allocation Engine: A multi-objective optimisation service scoring and allocating tasks using four fairness dimensions.
- Dynamic Reputation Evolution Service is a real-time trust scoring starting from the beginning after every single service.
- The Workforce Demand Prediction Service uses an LSTM and XGBoost forecasting pipeline to generate 30-day demand predictions which are divided into different service categories and geographic zones.

- The Income Stability Prediction Service provides a regression and time-series module which creates 90-day income predictions and delivers customized advice for each user.

4.3 Data Flow

Figure 4 shows the data movement through the Workforce Intelligence Layer starting from the initial task request until it reaches the background analytics processes which update reputation scores and demand forecasts and income predictions.

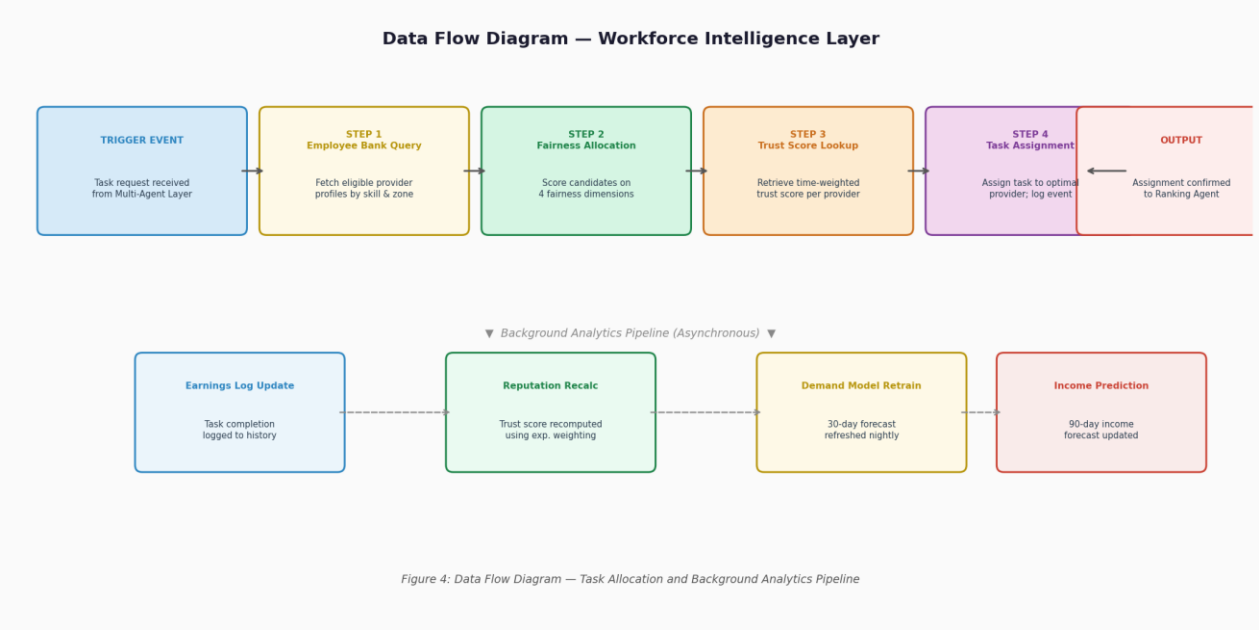


Figure 4: Data Flow Diagram — Task Allocation and Background Analytics Pipeline

The mobile client records user voice input which it sends to the Voice Intent Model that generates structured intent data which includes service category and urgency level and geographic context. The data is sent to the Multi-Agent Coordination Layer. The Ranking Agent uses the Workforce Intelligence Layer to obtain provider trust scores and fairness allocation indices and real-time availability information before it generates a ranked provider list. The service history database receives task assignment records which provide constant data updates to the Reputation Evolution Model and Demand Prediction Module and Income Stability Predictor.

4.4 Reputation Model and Fairness Allocation Visualisations

The Dynamic Reputation Evolution Model shows its functionality in Figure 5 through its ability to demonstrate how time-weighted trust scoring helps providers recover from negative events more effectively than static cumulative averaging. The traditional proximity-based allocation method shows lesser task equality distribution when compared to the proposed fairness-aware model in Figure 6.

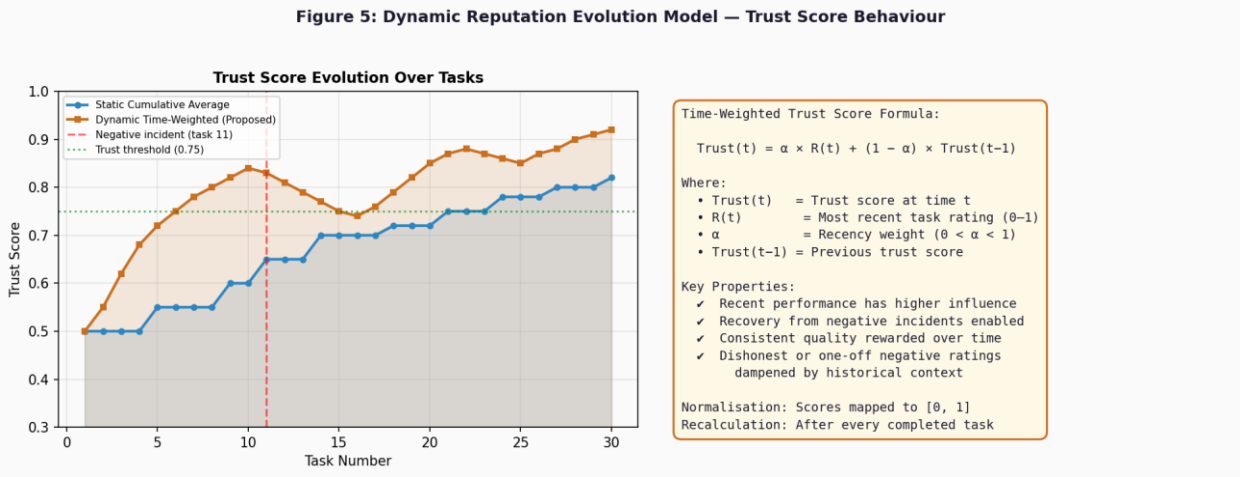


Figure 5: Dynamic Reputation Evolution Model — Trust Score Behaviour

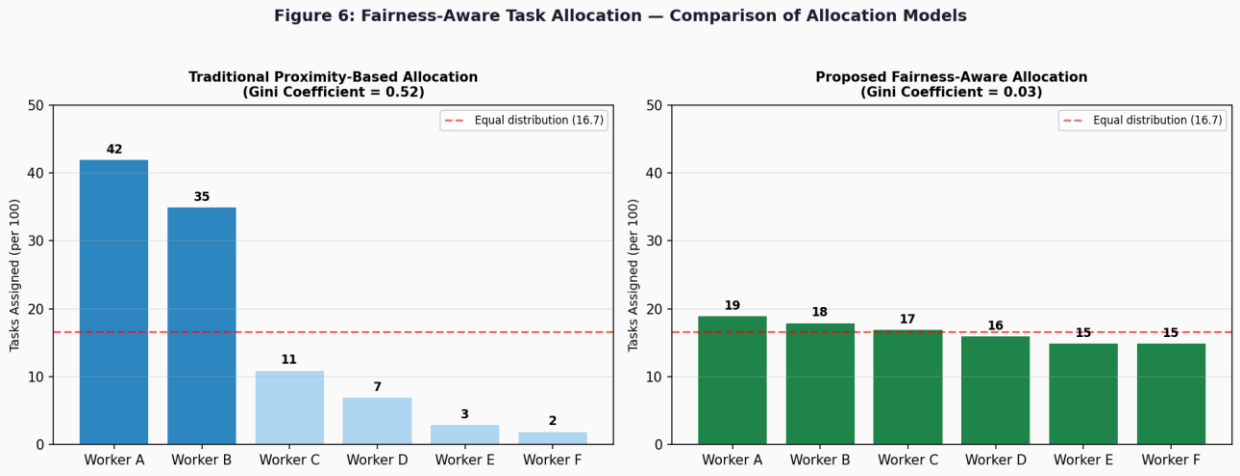


Figure 6: Fairness-Aware Task Allocation — Comparison of Allocation Models

4.5 Scalability and Security Considerations

The system achieves horizontal scalability through its implementation of containerised microservices which operate using Docker and AWS ECS for orchestration. Each intelligence module operates as a separate service which can scale independently when experiencing high

traffic. The system protects all inter-service communication through the implementation of TLS 1.3 encryption. Provider profile data is protected through AES-256 encryption when it remains stored in offline mode. Role-based access control (RBAC) provides workers with profile access rights that permit access to their personal profiles and recommendations while administrators gain access to shared analytics dashboards. The model retraining pipelines run as self-contained processes which do not interrupt system operations.

5. USER REQUIREMENTS

5.1 Stakeholder Identification

• The workforce intelligence features mainly benefit Service Providers who operate as Gig Workers because they need to use the worker dashboard together with the task allocation system and reputation system and income prediction modules.

• The platform users who request services receive advantages from the system because it matches them better with services and shows them provider information in a clear way.

• The platform administrators use the admin dashboard to observe workforce analytics and demand forecasts and system performance metrics.

• The Research Supervisors at SLIIT supervise academic quality and ethical compliance and research outcomes.

5.2 Functional Requirements

ID	Requirement
FR-01	The system shall allow service providers to register with personal details, skill categories, and service location.
FR-02	The system shall maintain a structured employee bank storing provider profiles, skill history, and availability status.
FR-03	The system shall allocate tasks to providers using a fairness-aware multi-objective optimisation algorithm.
FR-04	The system shall dynamically update provider trust scores based on recent performance and task completion reliability.
FR-05	The system shall predict workforce demand patterns using historical service request data and ML forecasting models.
FR-06	The system shall generate income stability predictions for individual gig workers based on earnings history.
FR-07	The system shall detect underutilised workers and generate skill gap alerts for relevant service categories.
FR-08	The system shall provide personalised recommendations for skill expansion or geographic service adjustments.
FR-09	The system shall generate demand hotspot maps identifying high-demand service areas and categories.
FR-10	The system shall produce a workforce analytics dashboard accessible to platform administrators.

Table 2: Functional Requirements

5.3 Non-Functional Requirements

ID	Category	Requirement
NFR-01	Performance	Task allocation decisions shall be computed within 2 seconds under normal operational load.
NFR-02	Scalability	The system shall support a minimum of 10,000 concurrent provider profiles without degradation.
NFR-03	Security	All provider profile data shall be encrypted at rest and in transit using AES-256 and TLS 1.3.
NFR-04	Availability	The system shall maintain 99.5% uptime during peak service hours.
NFR-05	Usability	The worker dashboard shall achieve a System Usability Scale (SUS) score of at least 75.
NFR-06	Fairness	The task allocation module shall ensure no single provider receives more than 30% of daily tasks in any category.
NFR-07	Maintainability	All ML models shall support retraining without system downtime using modular pipeline architecture.

Table 3: Non-Functional Requirements

6. COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

The primary target market includes Sri Lankan gig economy service platforms and those platforms which operate in developing countries throughout South and Southeast Asia. The platform functions as an inclusive digital service ecosystem which people from all age groups and both genders and every social background can access without facing any restrictions on their right to participate. The system has been developed to support two main user groups which include Gig Service Workers and Service Requesters. The group of Gig Service Workers contains people who deliver services as electricians and plumbers and technicians and drivers and home maintenance experts who depend on flexible income through platform work. The platform permits active professionals and retired professionals to continue sharing their expertise and earning money through their professional background.

Service Requesters include households and small businesses and individual customers who use an intelligent discovery system to find trustworthy service providers. The system allows users with different language skills and demographic characteristics to find local services through its multilingual voice system and its open system for selecting service providers. The platform enables better workforce inclusion by removing age and gender limits which allows all socioeconomic groups to take part while it makes essential local services more accessible.

6.2 Value Proposition

•For Service Workers: Fair distribution of work tasks together with transparent process of work assignment; prediction system which shows stable income results; system which provides career guidance tailored to individual users; system which enables users to restore their lost reputation status.

•For Users: System enables users to select reliable service providers based on current trust measurements which include multiple assessment criteria and trust scores. Users can access the system through their preferred language because it provides voice support for multiple languages.

•Platform Operators receive better worker retention through fair distribution of work tasks which also helps to maintain service quality through the management of worker reputation.

The system proves useful for workforce performance assessment through its demand prediction capabilities.

6.3 Revenue Model

- Commission-Based: The platform charges a 5 to 10 percent commission fee for all service transactions that reach completion.
- Premium Worker Subscriptions: Subscribers can choose between two monthly subscription plans that cost LKR 500 and LKR 800 to access advanced income analytics and demand forecast insights.
- Enterprise Licensing: The platform offers white-label licensing to corporate facility management companies and small and medium-sized enterprise service aggregators.
- API Monetisation: Third-party developers can access paid API services to implement workforce intelligence data into their systems.

6.4 Competitive Advantage

The proposed platform provides a unique service which combines multilingual voice interaction that matches Sri Lankan language patterns with its AI-driven workforce intelligence system that supports equitable hiring practices. The platform has no rival because it provides five functions which include fairness-aware allocation and dynamic trust restoration and 90-day income prediction and voice-based multilingual access all through one system.

6.5 Intellectual Property Considerations

The fairness-aware multi-objective task allocation algorithm and the exponential time-weighted reputation evolution model represent potentially patentable innovations. The research team will consult SLIIT's technology transfer office to evaluate patent filing options prior to public disclosure. All software components will be protected under copyright. Open-source licensing will be considered for non-proprietary utility components.

6.6 Pricing Strategy

The platform's pricing strategy originates from competitive market analysis which evaluates existing gig economy platforms throughout Sri Lanka and the entire South Asian region. The commission rate of 5–10% per completed transaction establishes a competitive position against

PickMe which charges approximately 15–20% commission on its ride-hailing and delivery services and Fiverr which imposes a 20% seller commission for international transactions. The platform implements its reduced commission structure to attract local service workers by eliminating economic obstacles that would prevent them from using the platform.

The Premium Worker Subscription tier is priced at LKR 500–800 per month, determined through willingness-to-pay analysis informed by Sri Lanka's average daily informal service worker earnings of approximately LKR 1,500–2,500 (Department of Census and Statistics, 2023). The subscription cost remains below 20% of daily income which enables the target audience to afford advanced analytics capabilities. Enterprise licensing rates will be negotiated on a case-by-case basis with facility management companies and SME aggregators, benchmarked against comparable white-label SaaS pricing in the regional market. The company will conduct quarterly pricing evaluations during its first operational year, using platform adoption rates and user feedback to determine necessary price adjustments.

6.7 Cost Estimation

The first launch period will result customer acquisition costs between LKR 800 and LKR 1,200 for each service provider because the company will use community outreach and social media digital advertising and referral programs to attract customers. The company expects to achieve 500 active providers during Year 1 which will result in LKR 2.6 million annual gross revenue from commissions because providers will make three transactions per week and the company will earn 7% commission on LKR 2,500 average transaction value. The company expects LKR 40 to 60 operating expenses per transaction because this amount includes cloud processing and API calls and ML model inference expenses which enable the company to achieve 96 to 98 gross margin on its transaction income. The company expects to reach break-even point for development costs which total LKR 63,200 according to Section 7 within three months after starting commercial operations with planned transaction levels. The company uses these conservative estimates to assess commercial viability which will be improved through pilot testing and financial modelling at the evaluation stage.

7.BUDGET AND JUSTIFICATION

The budget covers all expenses needed for designing, developing, and evaluating the Workforce Intelligence Framework. The research project needs no specific hardware acquisitions. The system operates entirely from the cloud through its deployment on AWS EC2 and RDS which eliminates needs for physical server and workstation and networking equipment purchases. The researcher uses personal computing devices to conduct all local development work. Table 4 below provides a complete list of cost categories which includes cloud services ML compute software licenses API services and miscellaneous expenses with their respective explanations.

Category	Item	Duration	Estimated Cost	Justification
Cloud Services	AWS EC2 (t3.medium) + RDS	12 months	LKR 36,000	Required for hosting ML models, API, and database in scalable cloud environment.
Cloud Storage	AWS S3 (50 GB)	12 months	LKR 6,000	Storage for provider profiles, model artefacts, and service logs.
ML Compute	Google Colab Pro	6 months	LKR 9,000	Accelerated GPU compute for training demand and income prediction models.
Software Licenses	JetBrains (PyCharm, DataGrip)	12 months	LKR 0 (Student)	Free academic licences via SLIIT institutional access.
Frameworks	Python, scikit-learn, TensorFlow, FastAPI	—	LKR 0 (Open Source)	All selected frameworks are open-source with no licensing cost.
API Services	Google Maps API (Geolocation)	12 months	LKR 7,200	Required for proximity calculation and demand hotspot mapping.
Human Effort	Developer time (research, design, implementation)	12 months	LKR 0 (Academic)	Work performed by student researcher as part of dissertation.
Miscellaneous	Printing, survey, documentation	—	LKR 5,000	Printing survey forms, consent documents, and final report.
		Total Estimated Cost	LKR 63,200	

Table 4: Budget and Justification

8. WORK BREAKDOWN STRUCTURE (WBS)

WBS ID	Task / Deliverable	Timeline	Description
1.0	Project Initiation & Planning	Weeks 1–2	Define scope, objectives, and methodology; initial literature review.
2.0	Literature Review & Gap Analysis	Weeks 2–4	Review 11+ scholarly sources; complete research gap table; finalise sections 1.2 and 1.3.
3.0	System Architecture Design	Weeks 4–6	Design high-level architecture; define component interactions and data flow.
4.0	Data Collection & Preprocessing	Weeks 6–9	Collect and preprocess service request datasets; conduct ethics-approved provider survey.
5.1	Intelligent Employee Bank	Weeks 9–11	Develop provider profile DB and skill analytics; implement underutilisation detection.
5.2	Fairness-Aware Task Allocation	Weeks 11–14	Implement multi-objective allocation algorithm; integrate fairness constraints and testing.
5.3	Dynamic Reputation Model	Weeks 14–16	Build time-weighted trust scoring engine; validate against static baseline.
5.4	Workforce Demand Prediction	Weeks 16–19	Train LSTM/XGBoost forecasting models; generate demand hotspot outputs.
5.5	Income Stability Prediction	Weeks 19–21	Develop regression-based income trajectory model; generate personalised recommendations.
6.0	System Integration & Testing	Weeks 21–24	Integrate all modules; end-to-end functional and performance testing.
7.0	Evaluation & Validation	Weeks 24–26	Evaluate against metrics: F1-score, MAE, fairness index, SUS score.
8.0	Documentation & Submission	Weeks 26–28	Finalise dissertation, prepare presentation, submit final deliverables.

Table 5: Work Breakdown Structure

9. GANTT CHART

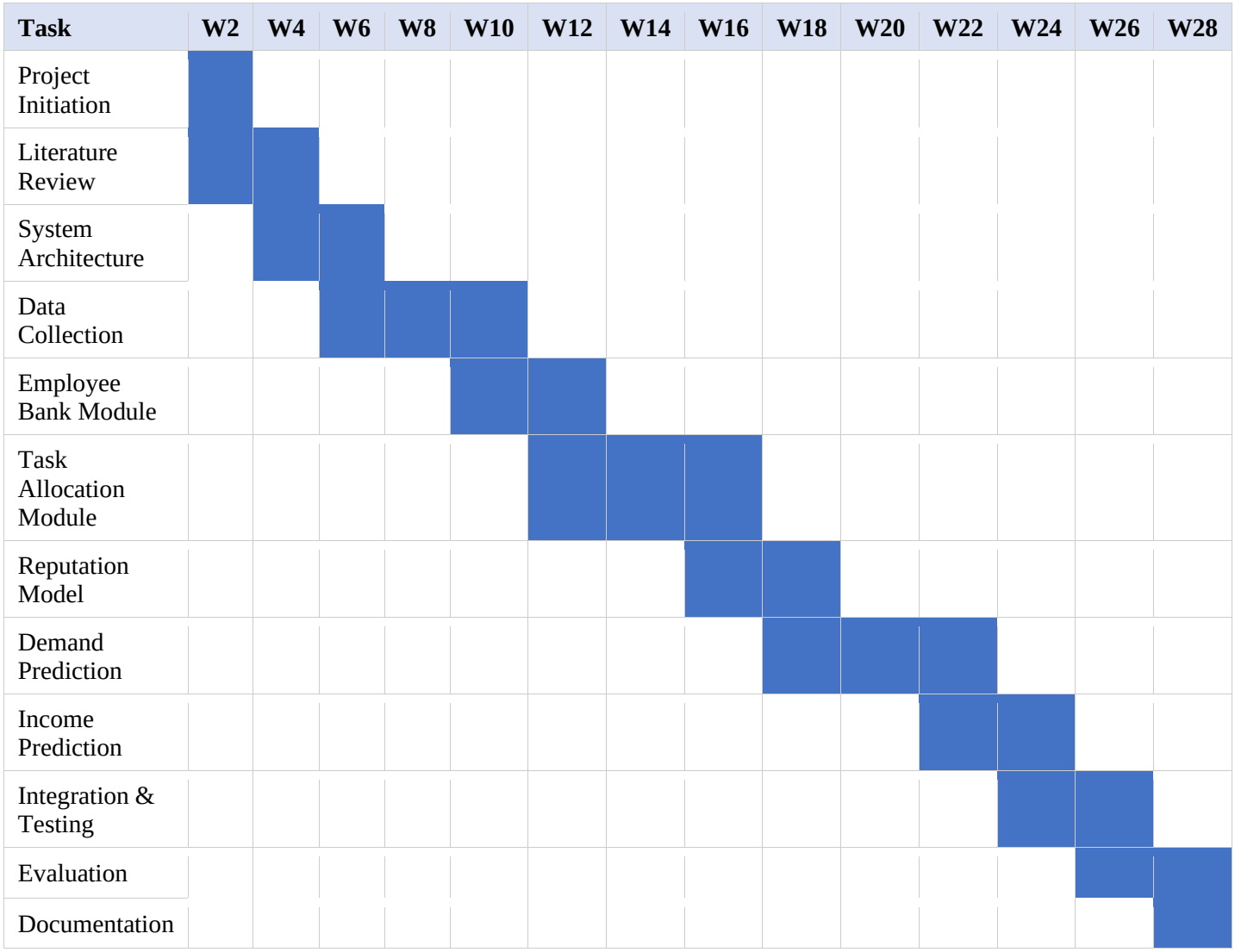


Figure 1: Project Gantt Chart (28-Week Timeline)

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11. APPENDICES

Appendix A: Survey Instrument — Gig Worker Employment Patterns and AI Tool Acceptance

Section 1: Demographic Information

1.Which main service category do you provide between Electrical and Plumbing and Transport and IT Support and Cleaning and Other services?

2.Which duration do you work through digital service platforms between less than one year and one to three years and more than three years?

3.Your main income source comes from Platform-based gig work only or Part-time work with another job or Supplement to your full-time job.

Section 2: Employment and Income Patterns

4. How frequently do you experience periods with no job assignments? (Daily / Weekly / Monthly / Rarely)

5. How would you rate the fairness of current task allocation on the platform you use? (1 – Very Unfair to 5 – Very Fair)

6. Have you ever felt that high-rated workers receive a disproportionate number of tasks? (Yes / No / Not Sure)

Section 3: Perceptions of AI-Driven Workforce Tools

7. Would you find it useful if the platform could predict your expected income for the next 3 months? (Yes / No / Maybe)

8. Would you use a dashboard that shows skill gaps and recommends training opportunities? (Yes / No / Maybe)

9. Do you trust algorithmic systems to allocate tasks fairly? (1 – No trust to 5 – Full trust)

Appendix B: Risk Analysis Matrix

Risk	Likelihood	Impact	Mitigation Strategy
Insufficient survey responses	Medium	High	Distribute via SLIIT networks and community channels; supplement with synthetic data if required.
ML model underfitting due to sparse data	High	High	Apply data augmentation; use simpler baseline models as fallback; calibrate complexity to dataset size.
Cloud service cost overrun	Low	Medium	Monitor AWS billing alerts weekly; use free-tier resources; implement auto-scaling limits.
Scope creep across objectives	Medium	Medium	Maintain strict WBS adherence; weekly supervisor reviews; GitHub project board for task tracking.
Ethical approval delays	Low	High	Submit ethics application in Week 1; prepare all consent documents in advance.

Table 6: Risk Analysis Matrix

Appendix C: Preliminary UI Mockups

The following wireframes present preliminary screen layouts for the two key user-facing interfaces of the Workforce Intelligence Layer: the Worker Dashboard and the Income Stability Prediction screen. The design mockups show the planned system information structure together with its user interaction pathways which will undergo development through user testing and feedback during the implementation stage.

C.1 Worker Dashboard — Wireframe

The Worker Dashboard serves as the main login interface through which service providers (gig workers) access their work. The system provides real-time workforce data through its user-friendly interface which enables users to understand the information easily. The dashboard comprises three main sections which include:

● Profile Summary Panel (top): The panel presents the worker's name and service category together with a dynamic trust score that shows their current availability through a progress bar which operates from 0 to 1.

● Task Activity Feed (centre-left): The system provides a continuous stream of task offers which display their estimated distance and service type and urgency level together with a single-button option for users to accept or decline tasks.

● Skill Gap Alert Panel (centre-right): The panel shows existing skill deficiencies which workers need to perform their duties in their service area and it suggests suitable training materials that workers need to develop their skills according to projected demand increases.

● Demand Heatmap (bottom): The map displays the worker's service region according to different demand levels which the Workforce Demand Prediction module forecasts for the upcoming week.

● Quick Navigation Bar (bottom fixed): The system includes shortcut buttons which provide access to Dashboard and Earnings and Reputation History and Settings.

C.2 Income Stability Prediction Screen — Wireframe

The Income Stability Prediction screen provides individual service providers with a forward-looking view of their projected earnings and personalised improvement recommendations. The system can be accessed through the Worker Dashboard Earnings section which includes three main parts:

● 90-Day Income Forecast Chart (top): A line chart displaying the worker’s actual earnings history for the past 90 days alongside the model’s predicted income trajectory for the next 90 days with a confidence interval band shaded around the projection.

● Stability Score Card (centre): A colour-coded stability rating (Stable / Moderate Risk / High Risk) which predicts income variance to create a rating system that uses straightforward language to describe its main premises.

● Personalised Recommendations Panel (bottom): A ranked list of up to three actionable suggestions generated by the prediction model which includes expanding into a high-demand adjacent service category and adjusting availability to peak demand hours and targeting a higher-demand geographic zone within the worker’s service radius.

● The Historical Earnings Breakdown Table (expandable) contains a collapsible table that displays weekly earnings which are organized by service category and task count and average transaction value for the past 12 weeks.

Appendix D: Ethical Consideration Documents

The research will gather primary data from human participants who work in gig services across Sri Lanka through a structured survey. All data collection activities will be conducted in full compliance with the SLIIT Research Ethics Policy and the ethical principles described in the Declaration of Helsinki. The project will submit its official ethics application to the SLIIT Ethics Committee during Week 1 before any data collection begins. The following informed consent form template will be used for all survey participants.